

Section: Electricity

Topic: Semiconductors

(a) Explain how the red LED operates

✓ 2 marks

✓ Answer:

- The **p-n junction** emits light when **electrons recombine** with **holes**
 - As electrons fall from the **conduction band** to the **valence band**, they release energy as a **photon**
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(b)(i) Determine the period of the a.c. supply

✓ 1 mark

Given:

- Each LED flashes **twice per second**

So:

- Flash rate = 2 Hz
- Period $T = \frac{1}{f} = \frac{1}{2} = 0.50 \text{ s}$

✓ Answer: 0.50 s

(b)(ii) Explain why red and blue LEDs do not light at the same time

✓ 2 marks

✓ Answer:

- The LEDs are **connected in opposite directions** in the a.c. circuit
 - In one half-cycle of the a.c. supply, only one type (red or blue) is **forward biased**, so it **lights up**
 - On the opposite half-cycle, the other LED lights up
- They are **on during alternate half-cycles**
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(b)(iii)(A) Determine the energy of a photon of wavelength 625 nm

✓ 2 marks

Use:

$$E = \frac{hc}{\lambda} \quad h = 6.63 \times 10^{-34}, c = 3.00 \times 10^8, \lambda = 625 \times 10^{-9}$$

$$E = \frac{6.63 \times 10^{-34} \times 3.00 \times 10^8}{625 \times 10^{-9}} = 3.18 \times 10^{-19} \text{ J}$$

✓ Answer: $3.18 \times 10^{-19} \text{ J}$

(b)(iii)(B) Explain difference between red and blue LED photon energies

✓ 2 marks

✓ Answer:

- **Blue photons** have **more energy** than red photons
 - This means blue LEDs have a **larger energy band gap** between conduction and valence bands
- A **greater energy transition** produces higher-energy (shorter wavelength) photons
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Revision Tips

- LEDs emit light when electrons drop across the band gap
 - Photon energy is **inversely proportional to wavelength**
 - A.C. alternates → only **one polarity** allows a specific LED to conduct
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